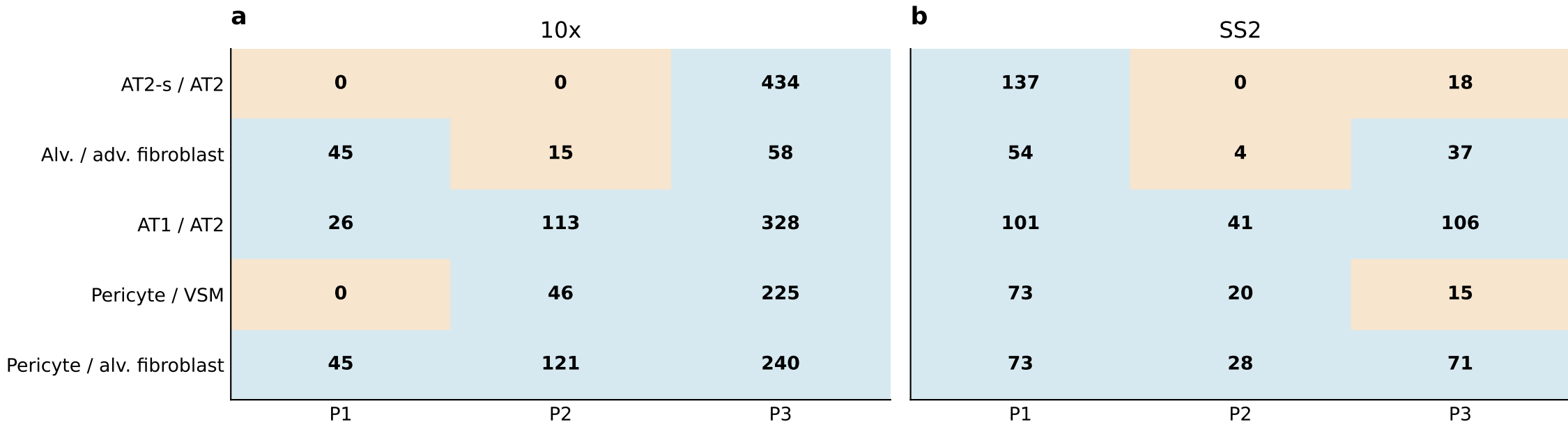


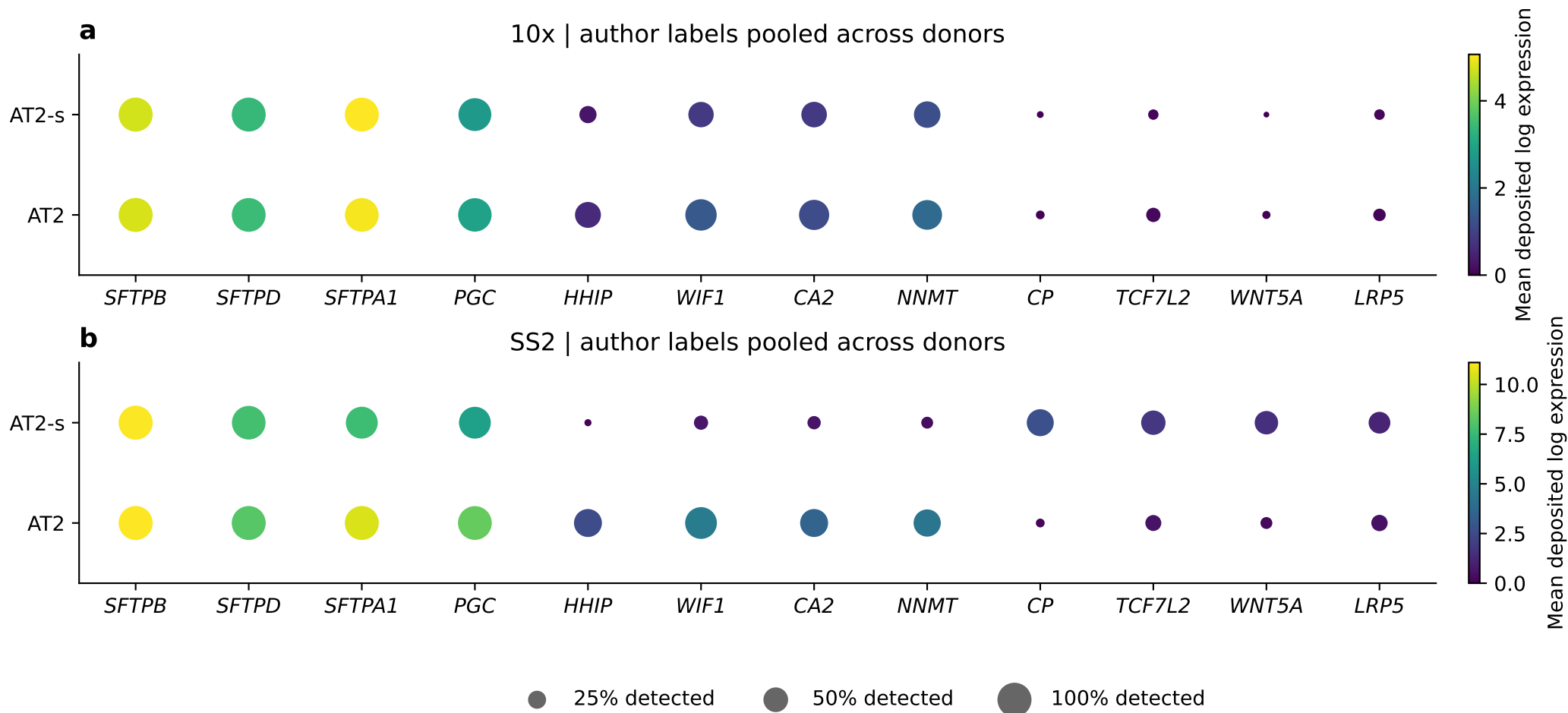
Matched distal-lung coverage limits the available comparisons



Numbers = cells in the smaller arm. Blue: at least 20 per arm; tan: below the fixed primary floor.

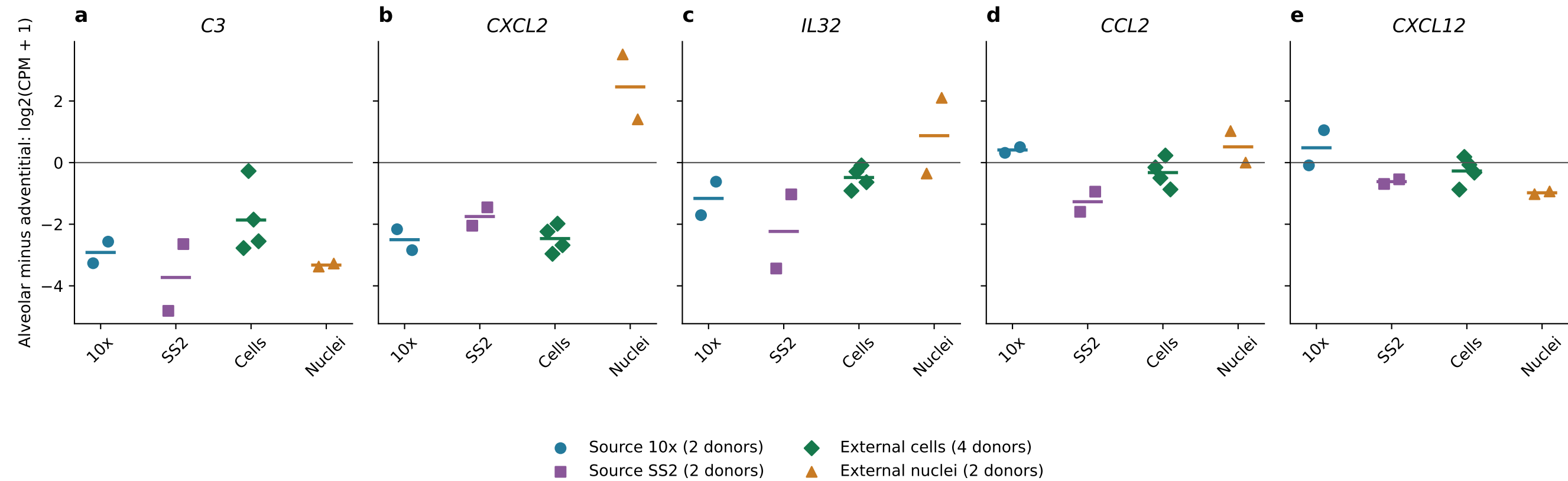
Donors remain separate; three donors are required for the planned donor-level description.

Targeted reconstruction of the published AT2 marker panel



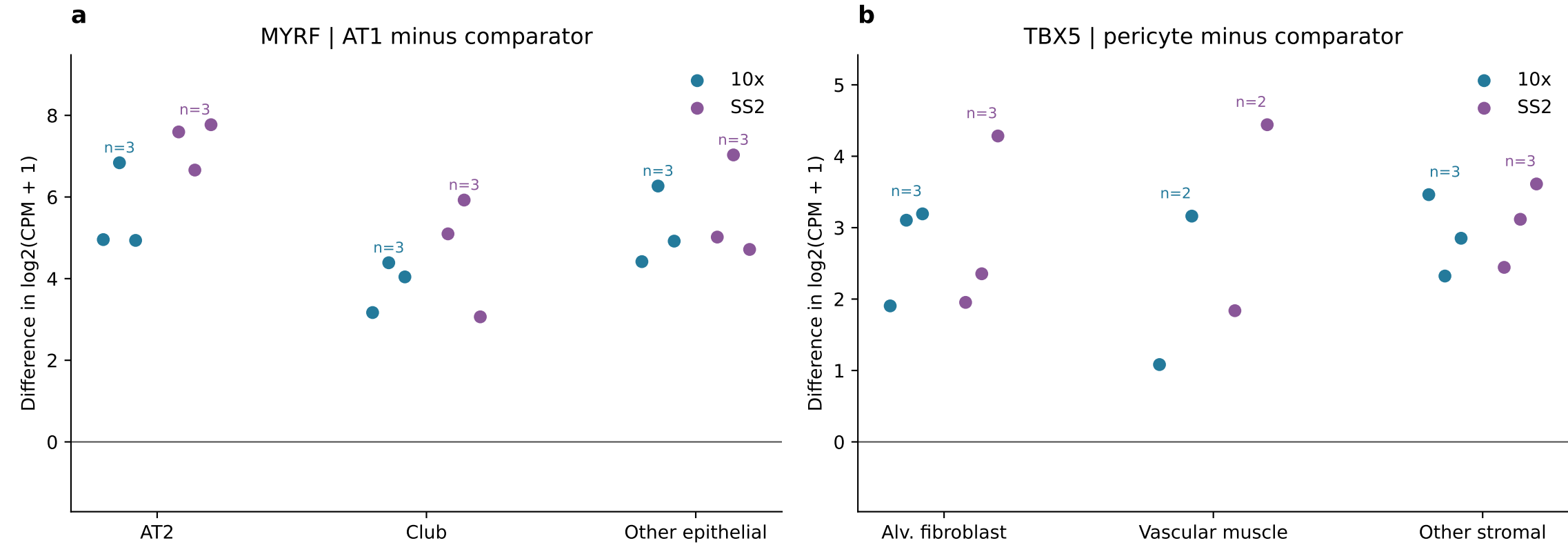
Dot area = detected fraction; colors use separate assay scales. Descriptive source reconstruction only.
 The source caption, colorbar and notebook disagree on assay; both curated matrices are displayed.

C3 transfers in direction; chemokine contrasts depend on sampling context



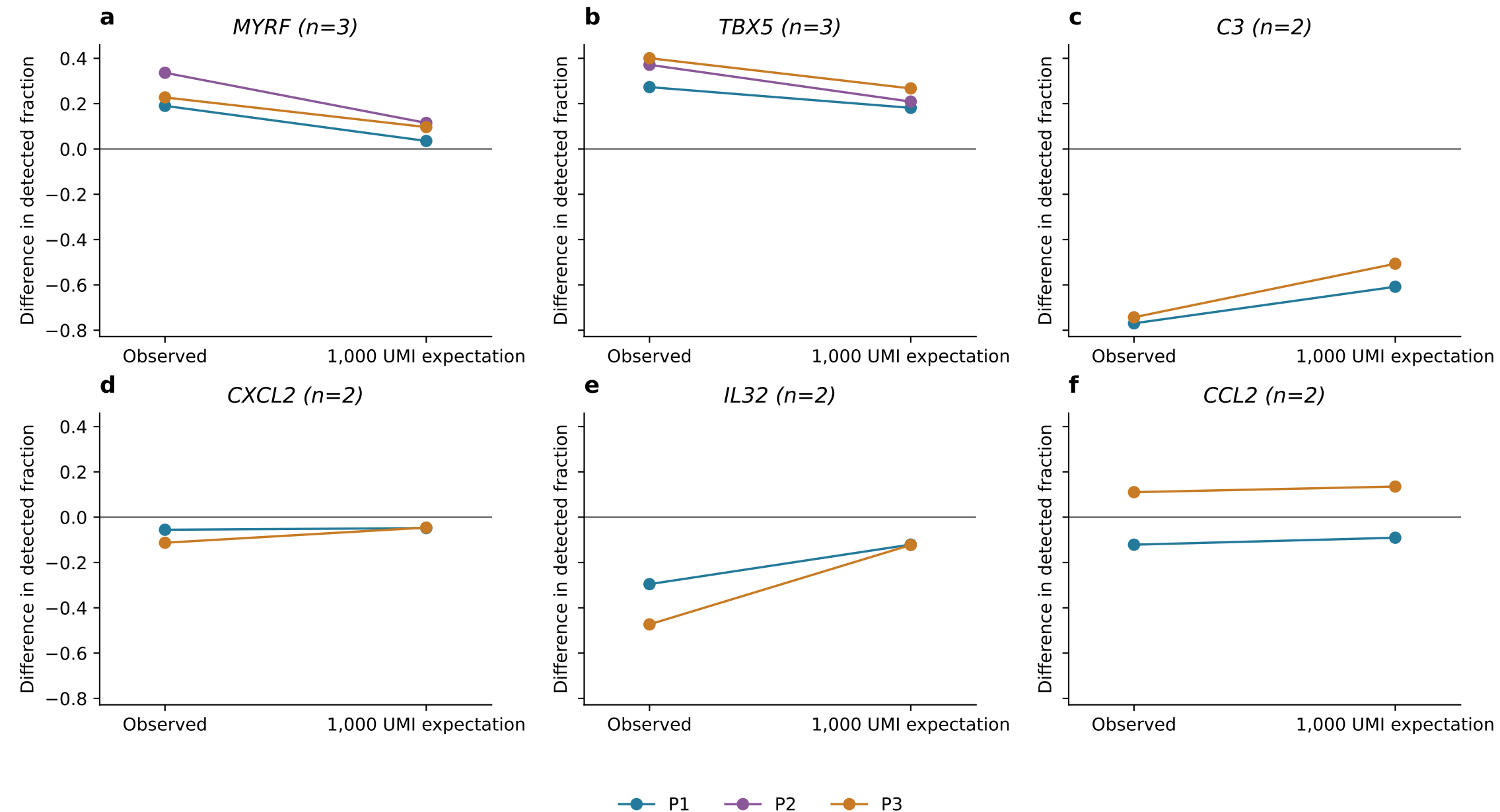
One point per donor; bars = donor means, not confidence intervals. External strata match location, protocol and assay. Cells and nuclei are separate: CXCL2 reverses in the nuclear subset. Lower values indicate adventitial enrichment.

Candidate regulator expression across comparator populations



Distal lung; minimum 20 cells in each arm; one point per donor. Author annotations are fixed.
Expression specificity supports reference use, not a demonstrated regulator function or repair outcome.

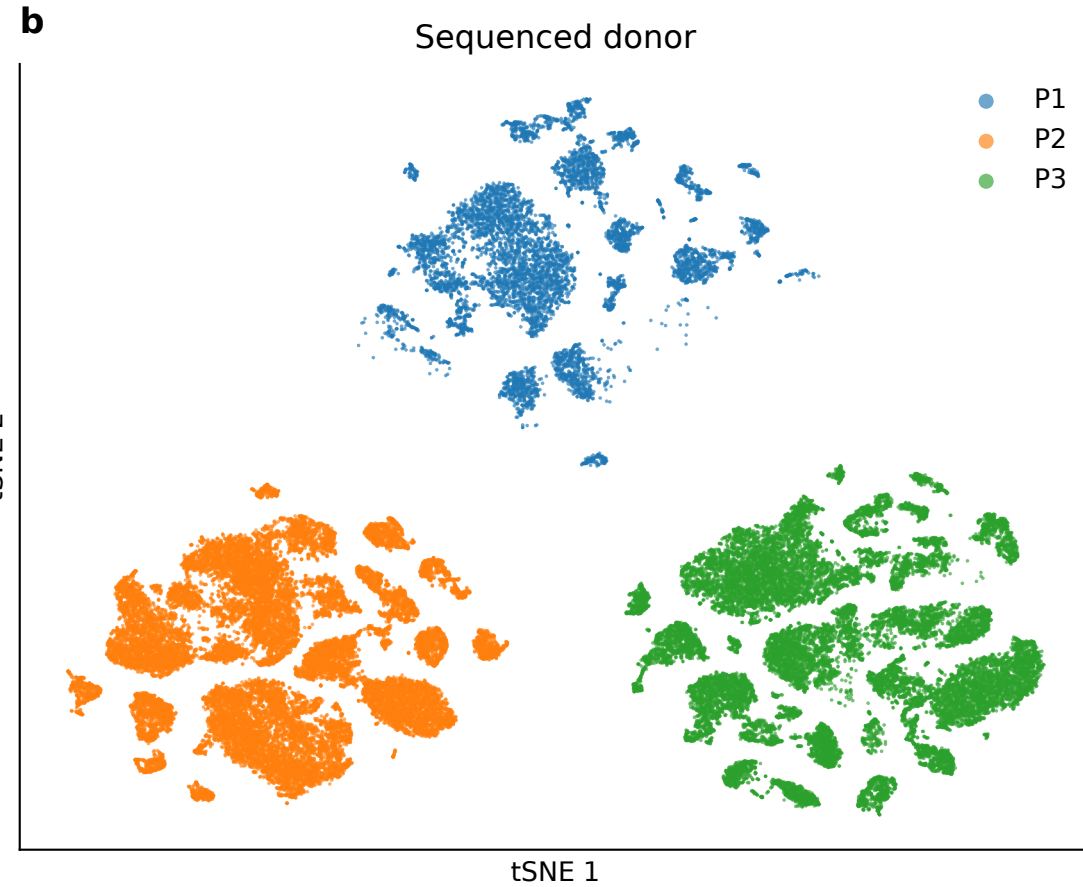
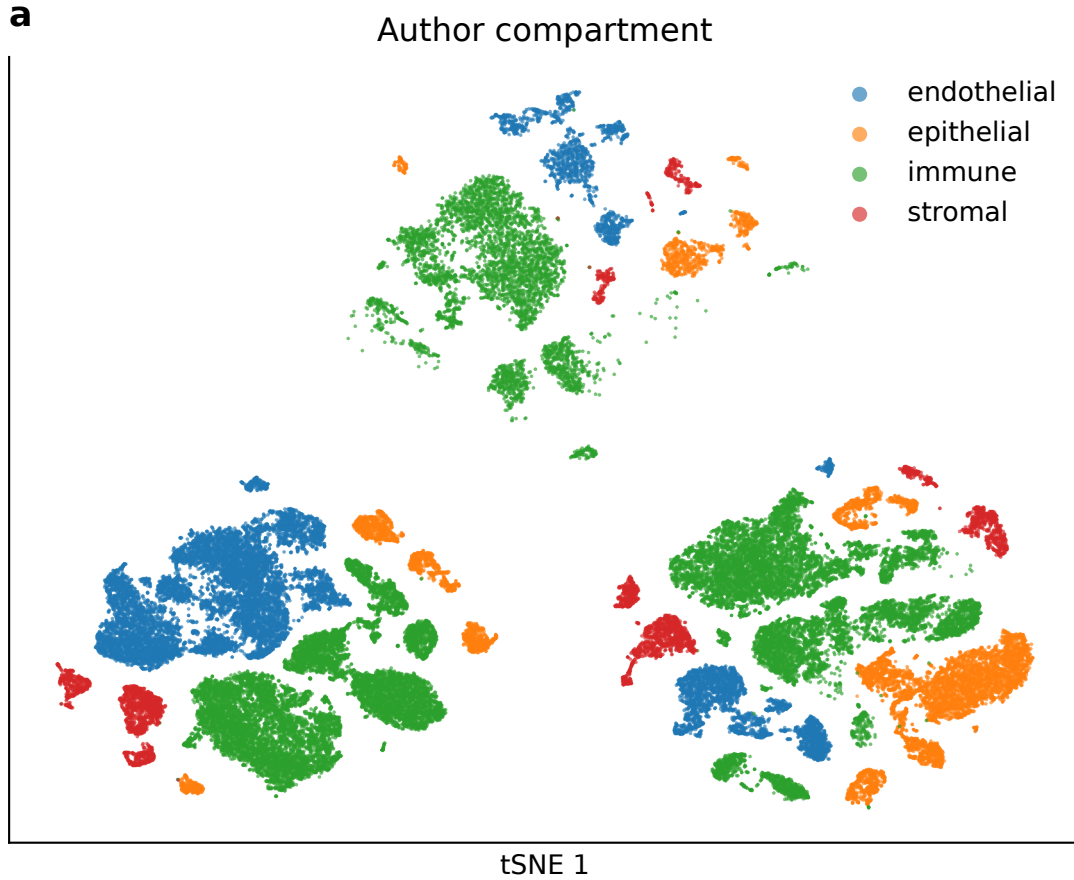
Within-donor detection contrasts after equal-depth sampling



10x distal lung; exact expectation from sampling 1,000 molecules without replacement. Lines connect the same donor.

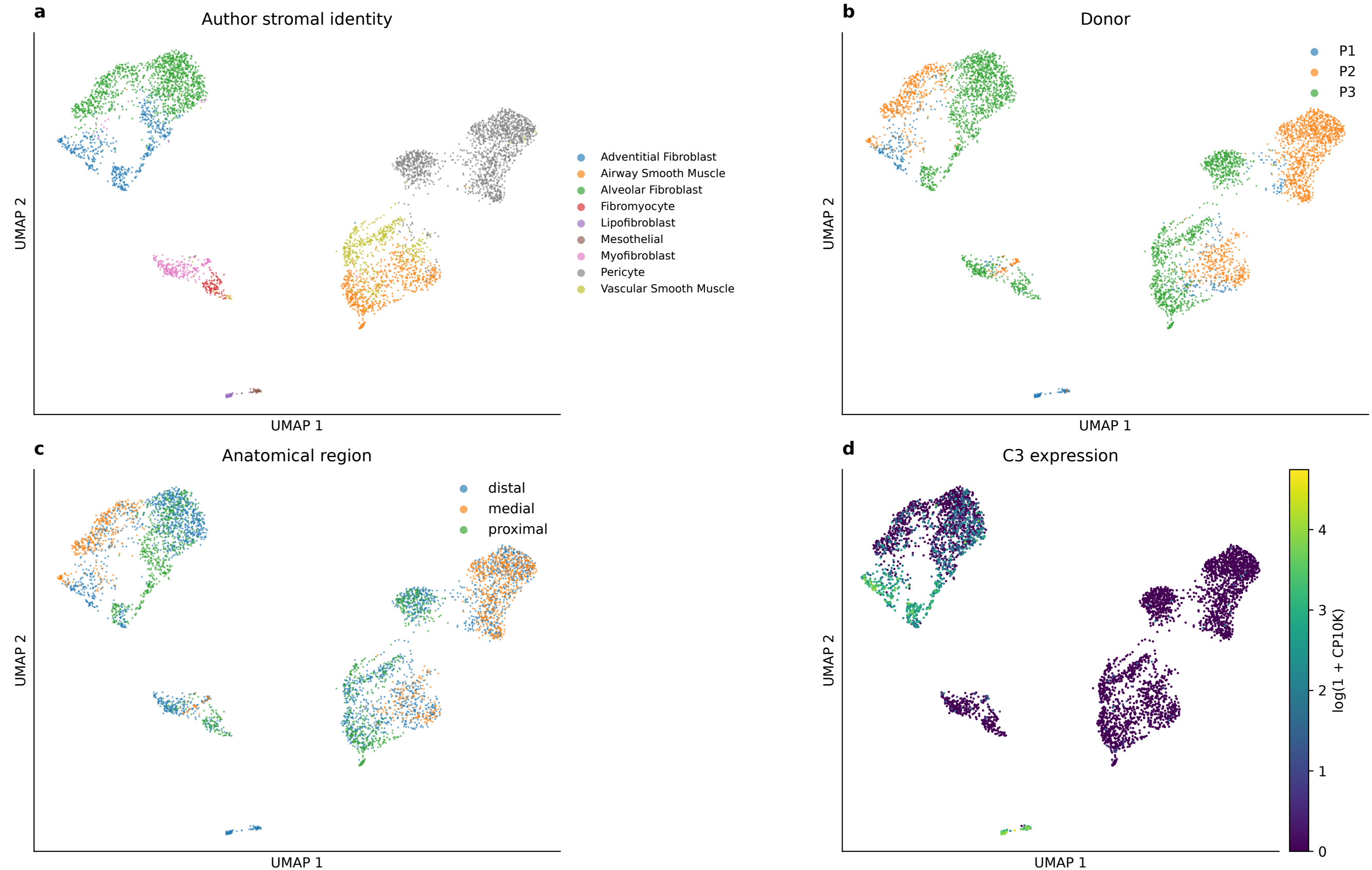
Contrasts: MYRF, AT1 – AT2; TBX5, pericyte – alveolar fibroblast; other genes, alveolar – adventitial fibroblast.

Deposited atlas t-SNE | 60,993 normal-lung 10x cells



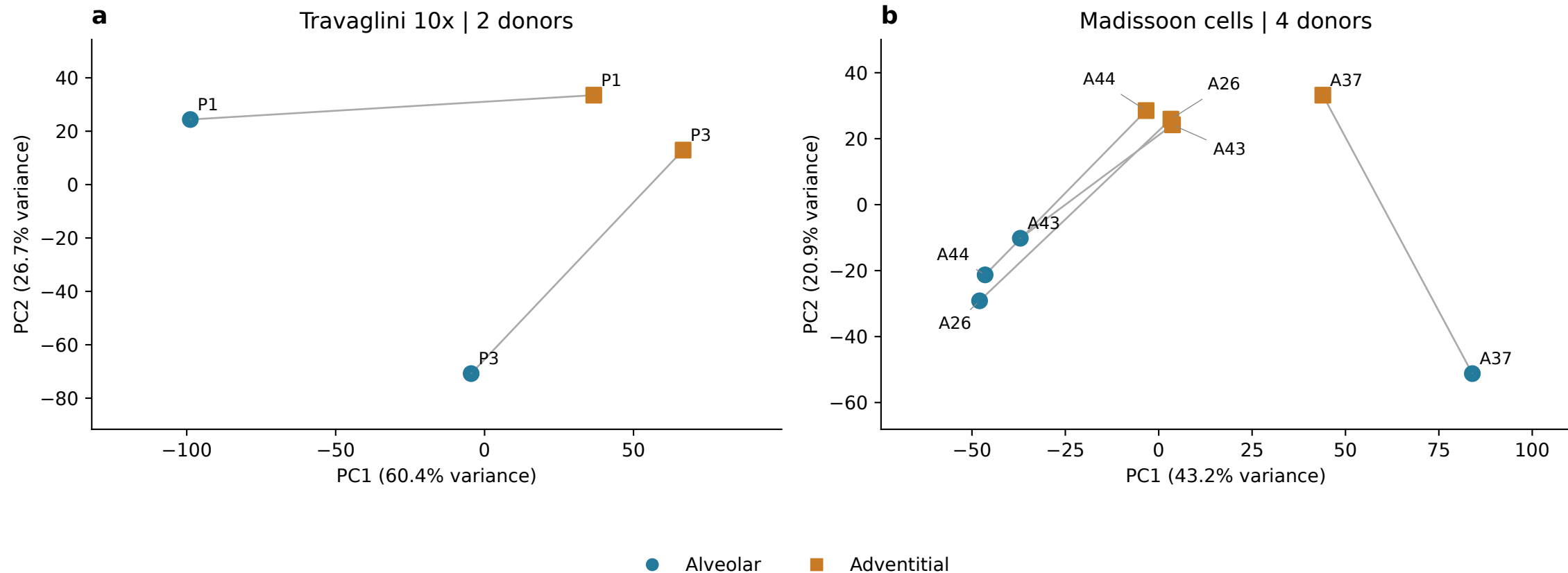
Original deposited coordinates, with blood excluded. Capture and sorting prevent interpreting point fractions as tissue prevalence.

New stromal UMAP | 5,033 cells, 1,500 variable genes, 30 PCs



Seed 20261002; 30 neighbours; minimum distance 0.4. Same coordinates in all panels; no batch correction or trajectory inference.
Donor and regional structure remain visible; author labels were not used to fit the embedding.

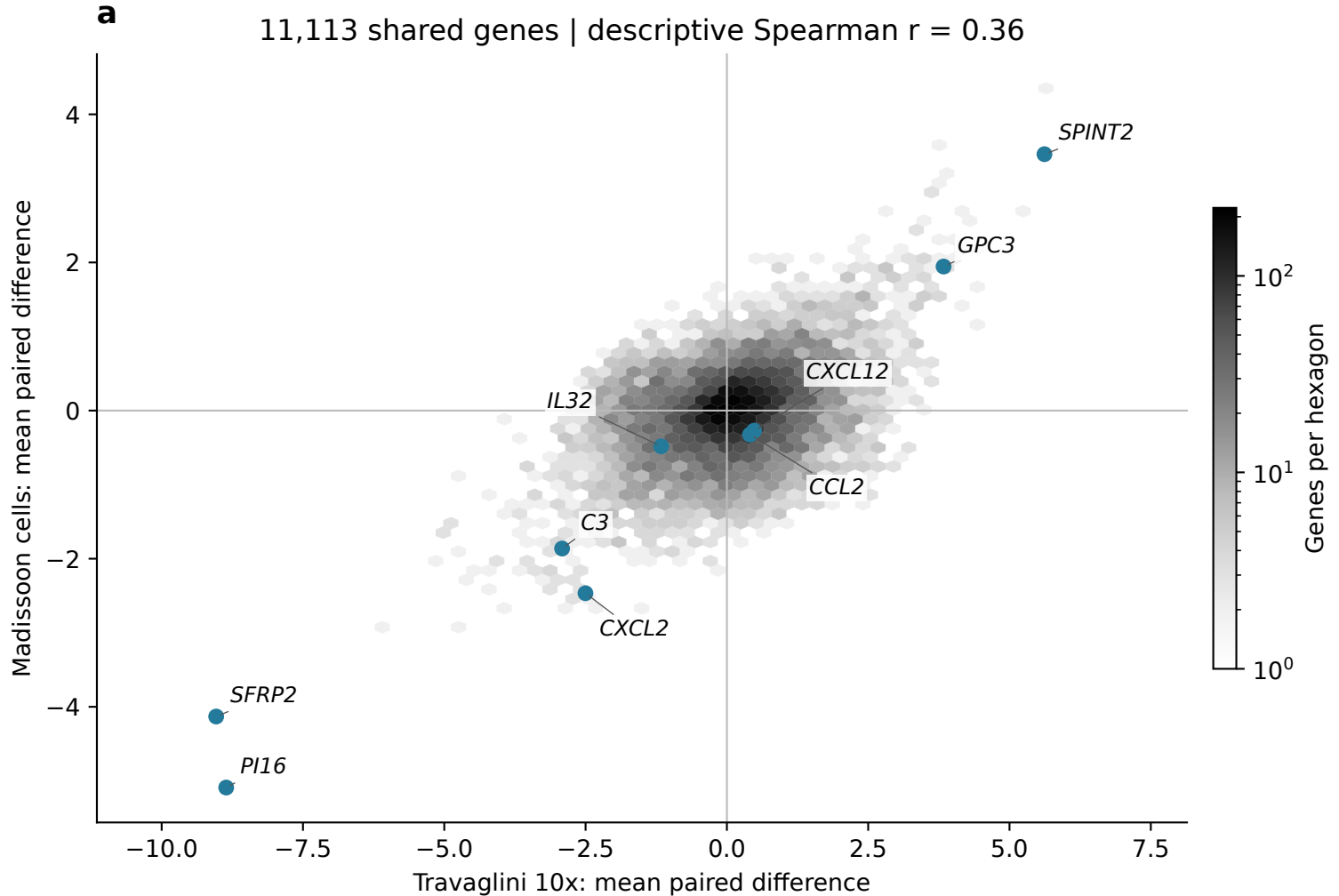
Donor-level fibroblast PCA in separate cohorts



Primary cell floor 20 per arm. Lines join the same donor; external profiles average eligible matched strata.

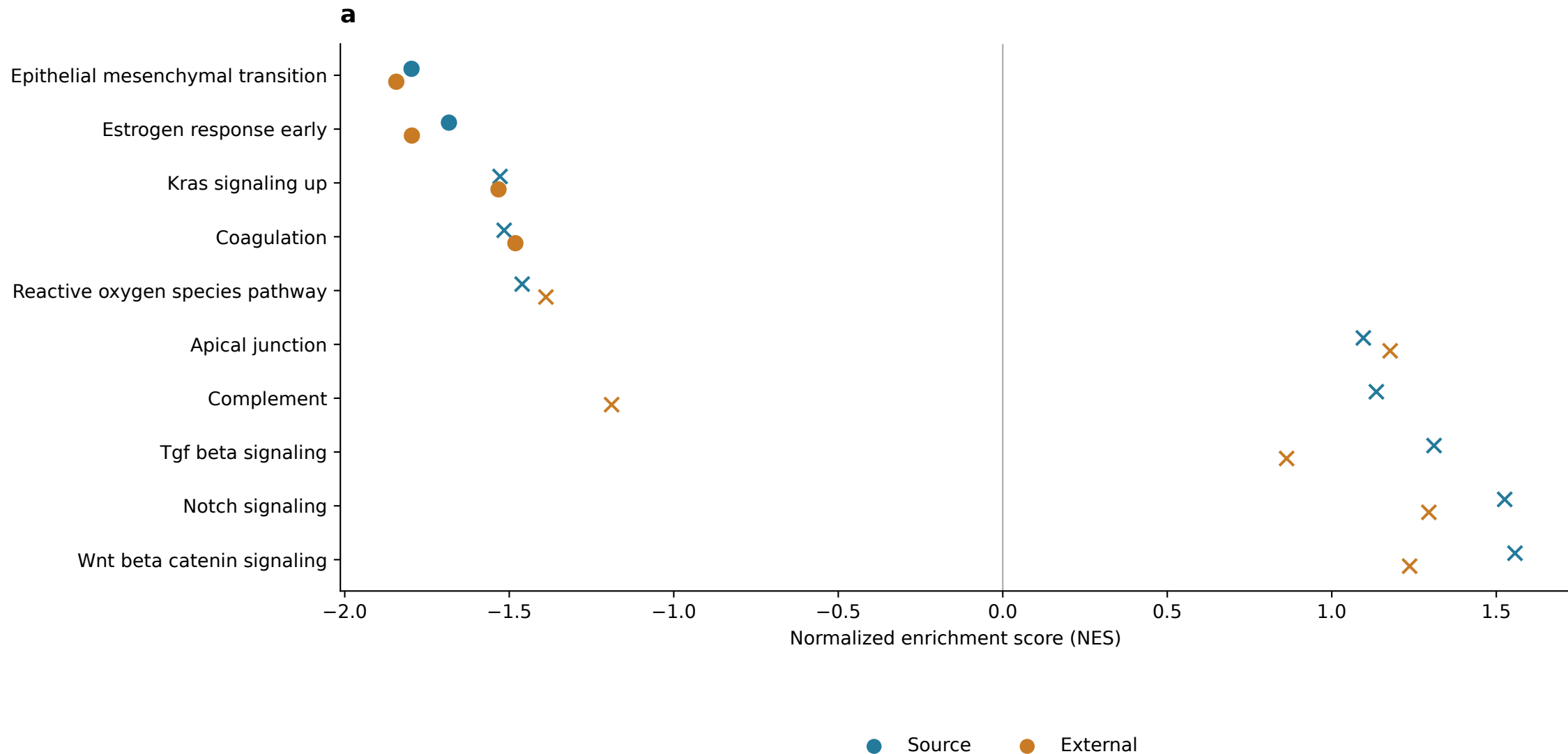
Top 2,000 variable genes per cohort within the shared expressed universe; axes are fitted separately.

Cross-cohort fibroblast expression contrasts



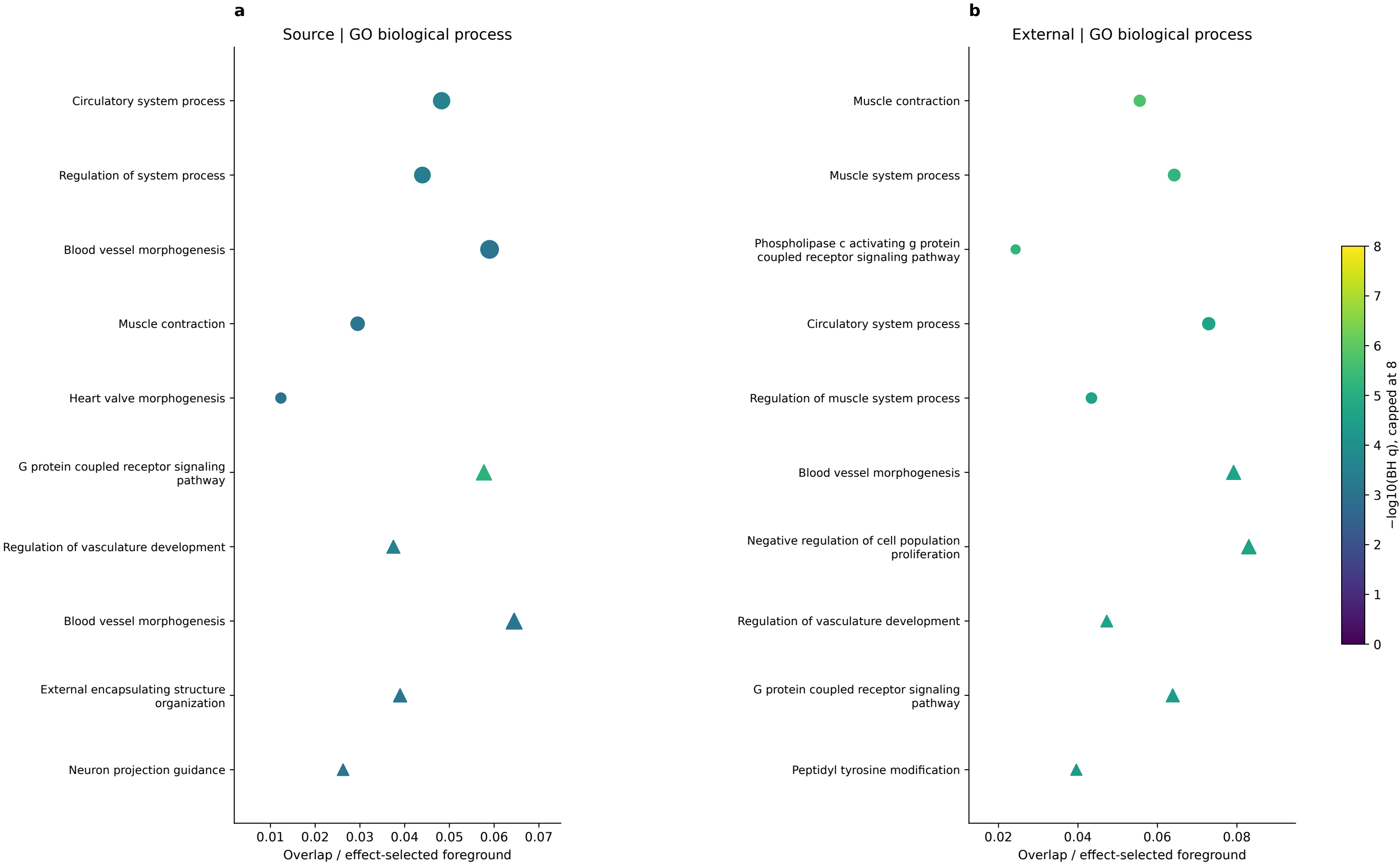
Alveolar – adventitial $\log_2(\text{CPM} + 1)$; source $n=2$ and external $n=4$ donors.
Highlighted genes were selected before this genome-wide fit. Correlation is across genes, not donors.

Hallmark GSEA of genome-wide fibroblast contrasts



Five strongest positive and negative source NES, with external estimates retained. Circles: gene-set FDR ≤ 0.05 ; crosses: > 0.05 . 1,000 gene-set permutations; these q-values are not donor-level biological significance tests. All eligible sets are tabulated.

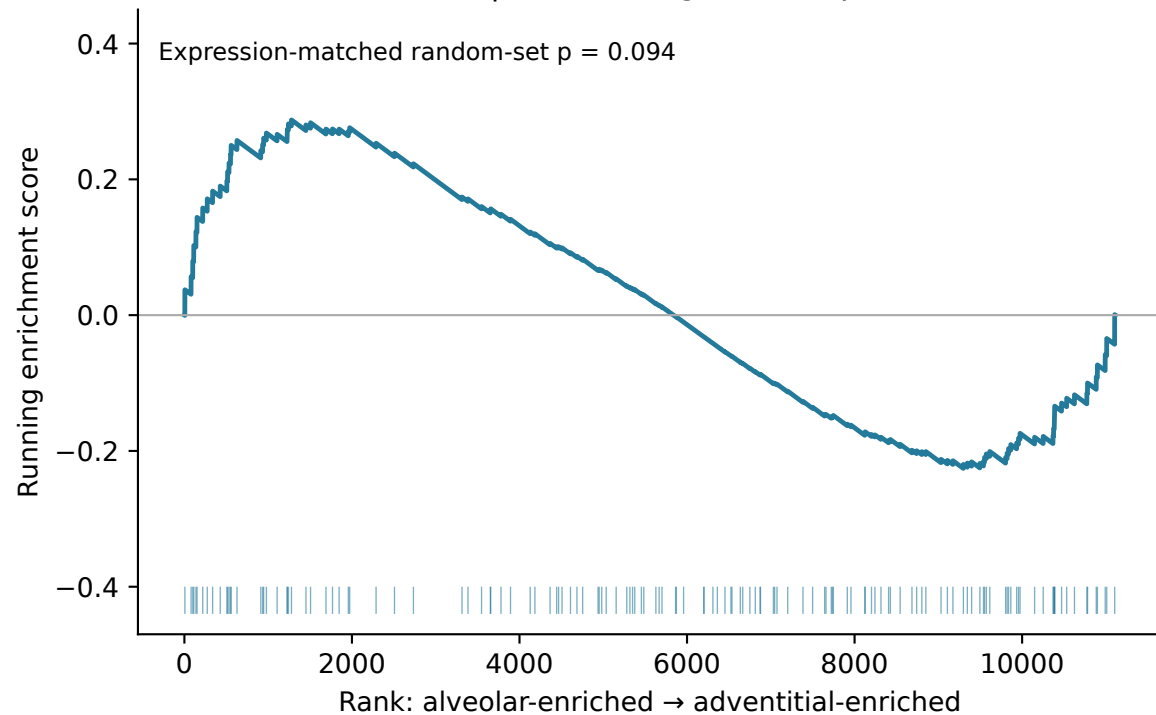
GO annotation of consistent, effect-selected fibroblast genes



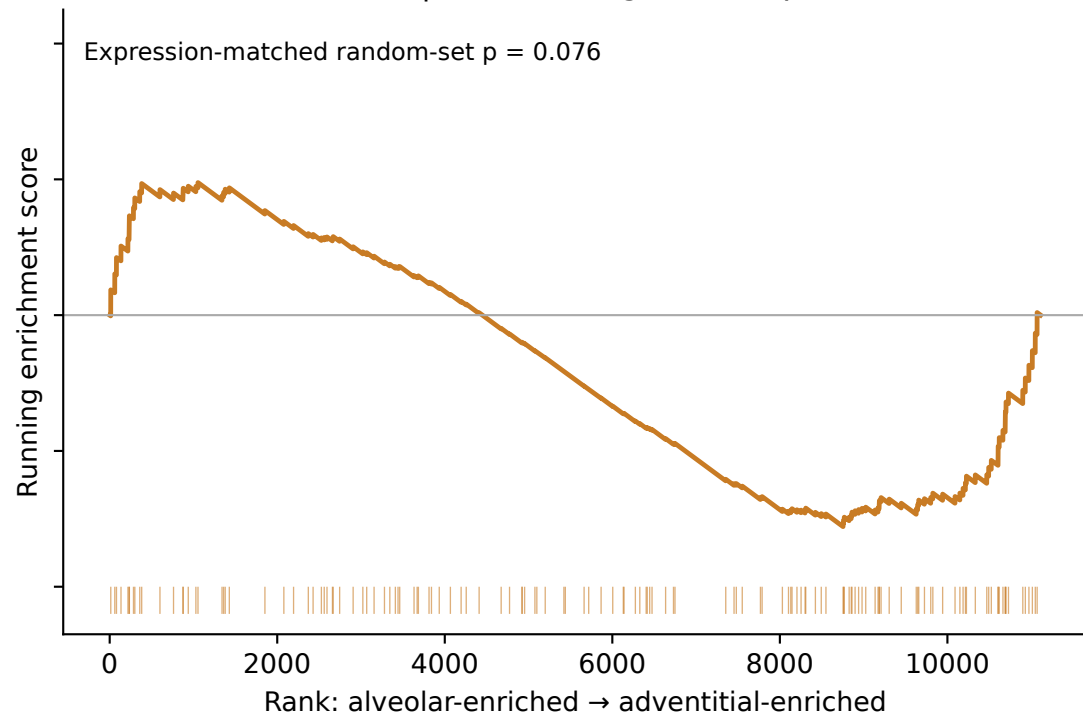
Circles: alveolar-up; triangles: adventitial-up; area scales with overlap count. Top five terms per direction/cohort; related GO terms overlap.
Foreground: $|\text{mean paired difference}| \geq 1$, same sign in 2/2 source or $\geq 3/4$ external donors. Background: 11,113 expressed genes.
BH correction covers all eligible terms and both directions within cohort. This is annotation enrichment, not a DE-gene significance test.

C3 direction does not generalize to the entire complement gene set

a Source | NES 1.14; gene-set q 1.00



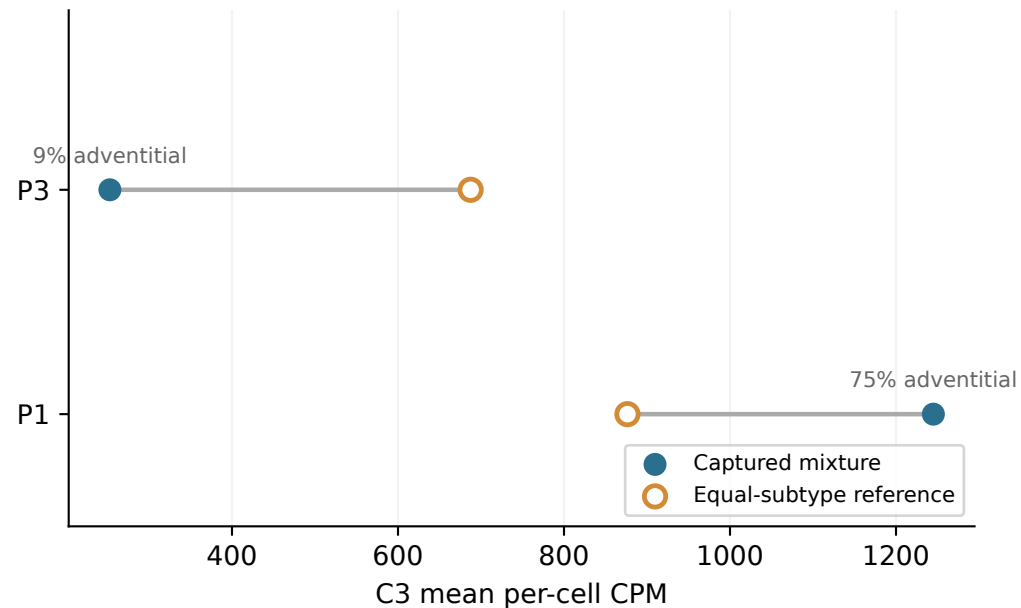
b External | NES -1.19; gene-set q 0.32



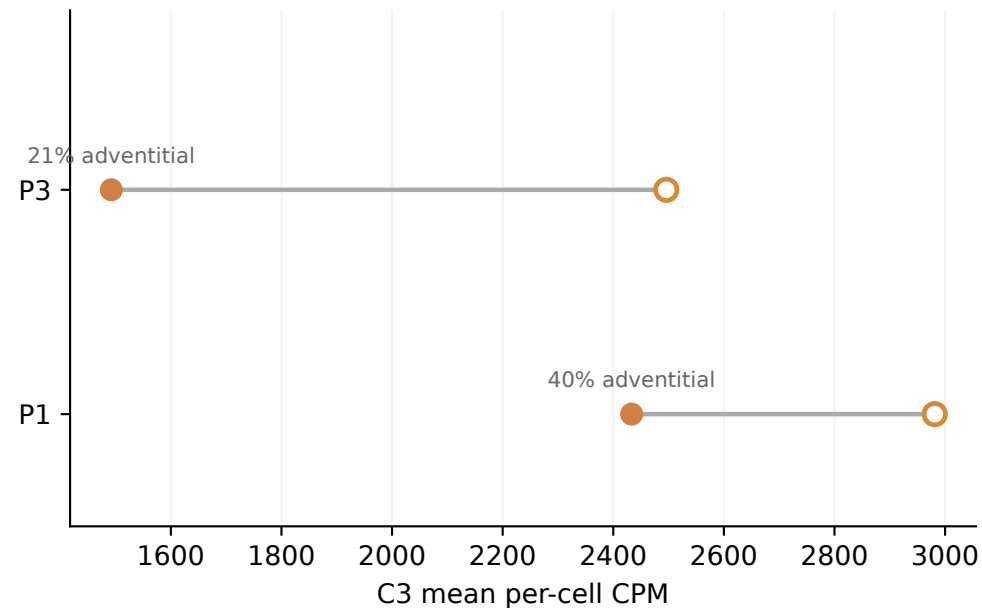
HALLMARK_COMPLEMENT selected before enrichment; 141 genes overlap the common universe. Ticks show set-member ranks.
Opposite enrichment directions and non-small adjusted q-values retain the whole-program claim on hold.

a

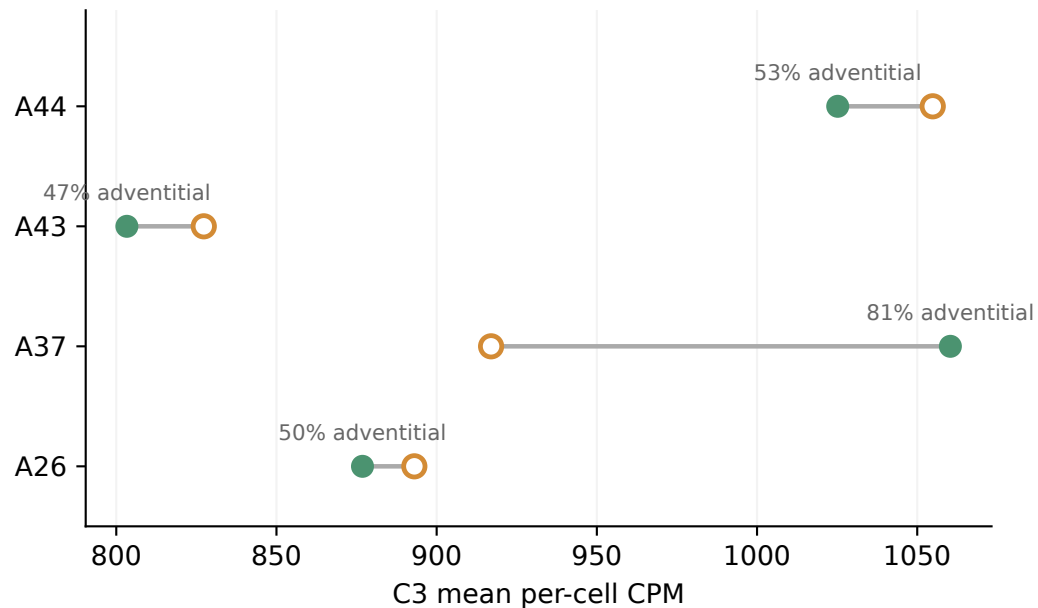
10x

**b**

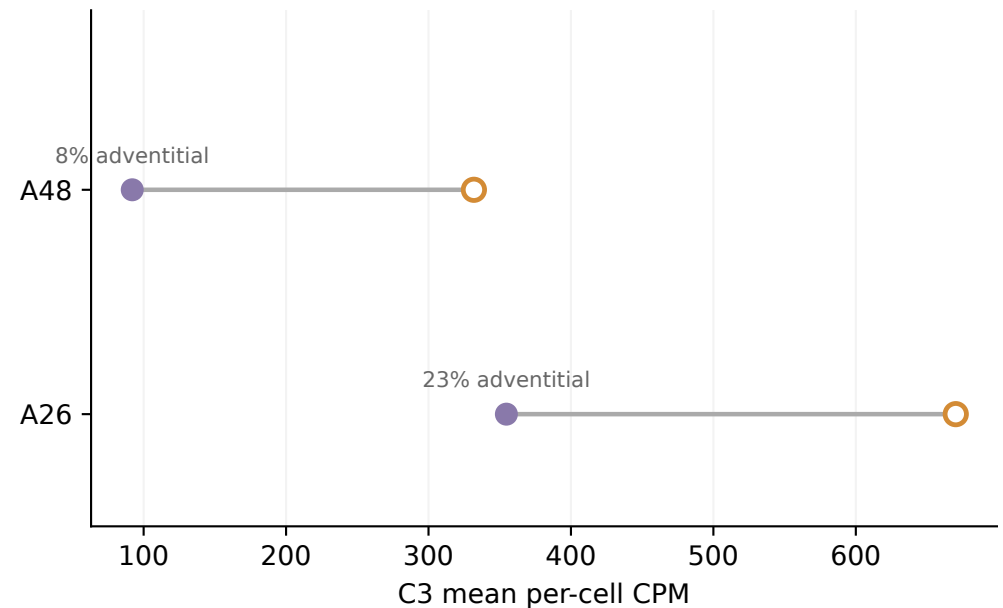
SS2

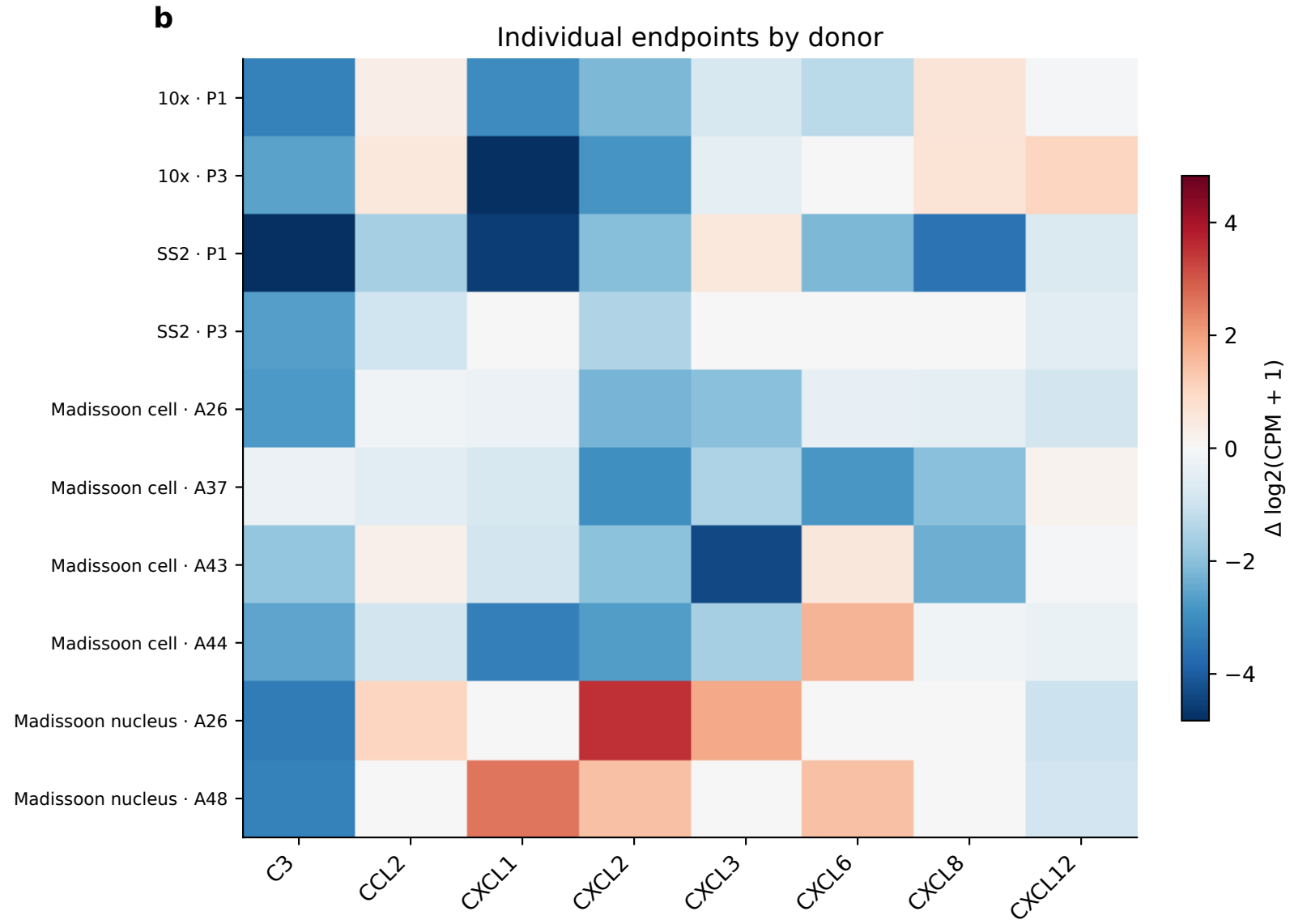
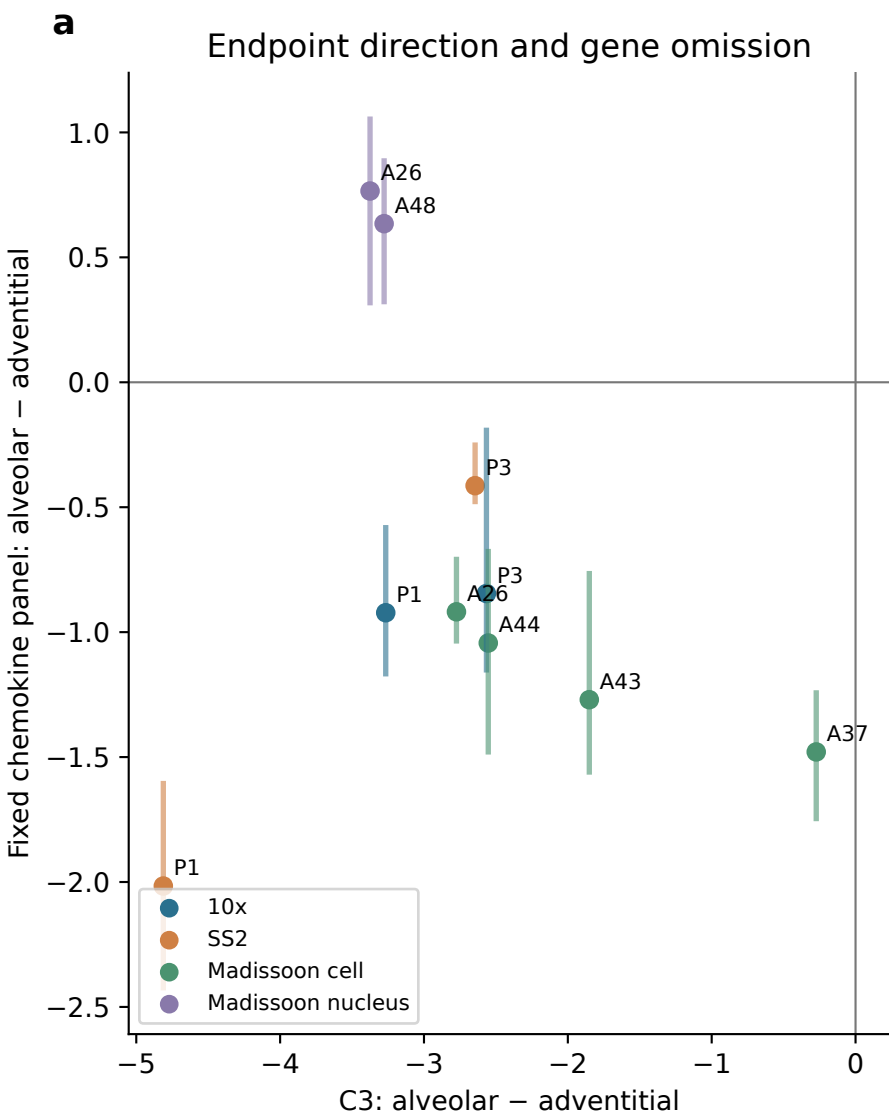
**c**

Madisson cell

**d**

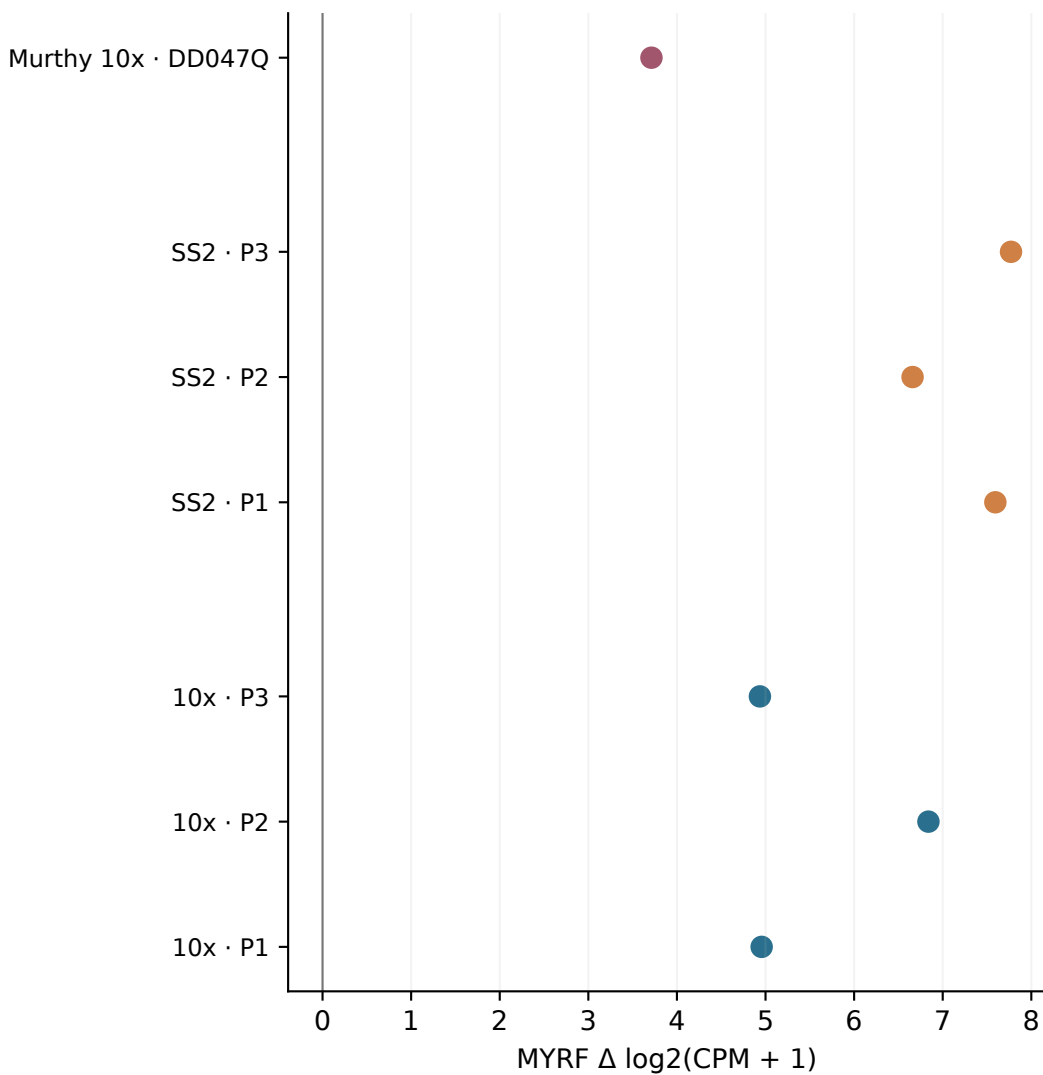
Madisson nucleus





a

AT1 versus AT2

**b**

AT1 versus mesothelium

